

ZAALIMA DEVELOPMENT INTERNSHIP

WEEK 4 PROJECT REPORT

Student Name: Pooja M

Project: Customer Churn Prediction & Lifetime Value (LTV) Engine

Internship: Zaalima Development

Duration: Week 4

PROJECT OVERVIEW

The objective of Week 4 was to build interactive dashboards for customer churn and lifetime value analysis, connect the visualization platform with the backend database and API, and prepare the application for deployment. This phase focused on transforming analytical results into actionable business insights through user-friendly dashboards.

TOOLS AND TECHNOLOGIES USED

- Python
- PostgreSQL
- FastAPI
- Metabase
- Docker
- Pandas
- Scikit-Learn
- GitHub

DATABASE AND API INTEGRATION

Connected Metabase to the PostgreSQL database containing customer records and prediction outputs. Integrated FastAPI endpoints to fetch churn prediction and LTV results dynamically for dashboard visualization.

DASHBOARD DEVELOPMENT

Created interactive dashboards to visualize overall customer churn distribution, high-risk churn customer segments, Customer Lifetime Value categories, contract type analysis, monthly charge trends, and customer tenure distribution.

LTV SEGMENTATION

Customers were categorized into High Value Customers, Medium Value Customers, and Low Value Customers based on predicted lifetime value. This segmentation helps businesses prioritize retention efforts and optimize marketing investments.

DEPLOYMENT PREPARATION

Containerized the application using Docker to ensure portability and simplified deployment across different environments. Verified application functionality, dependencies, and configuration settings.

DOCUMENTATION

Prepared technical documentation covering project architecture, database schema, API endpoints, machine learning workflow, and dashboard usage guide.

KEY ACHIEVEMENTS

- Successfully connected Metabase with PostgreSQL.
- Built interactive churn and LTV dashboards.
- Implemented customer segmentation based on predicted value.
- Containerized the project using Docker.
- Completed deployment-ready documentation.

OUTCOME

Successfully completed the final phase of the Customer Churn Prediction & Lifetime Value Engine project. The system now provides predictive insights, interactive business dashboards, API-based access to predictions, and a deployment-ready architecture for real-world business use.